SMART STREET LIGHT TECHNOLOGY USING OCCUPANCY SENSOR

A project report in partial fulfilment of the requirement for the award of

Bachelor of Technology

in

Electrical Engineering

Submitted by:

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Certificate of Approval

This is to certify that the project report entitled "Smart Streetlight technology using occupancy sensor" being submitted by Dhrubajyoti Sonowal (D/21/EE/002), Tezenmo Woch (D/21/EE/012), Nending Tagia (D/20/EE/108) and Vekhoyi Nyekha (D/21/EE/113) in partial fulfilment for the award of the Degree of Bachelor of Technology in Electrical Engineering of the North Eastern Regional Institute of Science and Technology (NERIST) is based on the theoretical and experimental studies carried out by them during the academic semesters, January-May, 2025 in a manner satisfactory to warrant its acceptance as a pre-requisite for the award of the Degree in Electrical Engineering of the Institute.

It is understood by this approval that the undersigned do not endorse or approve any statement made, opinions expressed, or conclusion drawn therein but only for the purpose for which it has been submitted

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DECLARATION

We hereby declare that the project entitled "Smart Streetlight technology using occupancy sensor" is a prerequisite project work carried out by us towards partial fulfilment of the award of Degree in Electrical Engineering submitted to the Department of Electrical Engineering, **North Eastern Regional Institute of Science and Technology**, **Nirjuli, Arunachal Pradesh** under the guidance of Dr Piyali Das, Assistant Professor, Dept. of Electrical Engineering during the academic semester, Jan-May, 2025**.**

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ABSTRACT

This project focuses on the development of an occupancy sensor and studying the possible energy savings that can be done within NERIST by incorporating the said occupancy sensor in the streetlights. We begin by measuring the light intensity of a few streetlights in certain areas of NERIST by using a luxmeter to establish a baseline for evaluating future improvements. Accurate light intensity measurements are crucial for assessing the efficiency and performance of street lighting systems, particularly in urban environments where safety and visibility are paramount. The collected data will serve as a reference point for comparing the performance of streetlights after the integration of occupancy sensors.

Occupancy sensors will enable the streetlights to adjust their brightness dynamically based on pedestrian or vehicular presence, optimizing energy consumption while maintaining safety. This report outlines the methodology for data collection, the initial results, and the importance of using luxmeter data to benchmark energy efficiency and illumination effectiveness. By comparing the pre- and post-installation data, the project aims to demonstrate the impact of smart lighting technologies on reducing energy wastage and enhancing sustainability in urban infrastructure

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List of Abbreviations

Abbreviations	Full form
IC	Integrated Circuits
LED	Light Emitting Diode
IR	Infrared
RL	Relay
TRIAC	Triode for Alternating Current
SPDT	Single Pole Double Throw
DIAC	Diode for Alternating Current

List of Symbols

Symbol	Description	Unit
V	Voltage	Volts (V)
C	Capacitor	Farad(F)
R	Resistance	Ohms (Ω)
P	Power	Watts (W)
f	Frequency	Hertz (Hz)
I	Current	Amperes (A)

CHAPTER 1: INTRODUCTION

Street lighting plays a crucial role in urban infrastructure by ensuring safety and security in public spaces. However, traditional street lighting systems operate continuously throughout the night, leading to significant energy wastage and increased operational costs. With growing concerns over energy efficiency and environmental sustainability, there is a pressing need to develop smarter, more efficient lighting solutions.

Modern street lighting systems employ energy-efficient technologies, such as LED (Light Emitting Diode) lights, which offer long service life, low energy consumption, and superior illumination. These systems are often integrated with smart technologies, enabling features like automated dimming, remote monitoring, and fault detection, which enhance operational efficiency and reduce maintenance costs. Street lighting projects must consider factors such as the appropriate spacing of light poles, the height and angle of installation, and the selection of luminaires to ensure uniform illumination without glare. Furthermore, the integration of renewable energy sources, like solar-powered streetlights, is gaining popularity, aligning with global sustainability goals. Proper planning and periodic maintenance of street lighting infrastructure are essential to ensure reliability and cost-effectiveness, making it a vital component of urban planning and infrastructure development.

The power consumption of street lighting systems varies significantly depending on the type and scale of the lighting infrastructure, as well as the technology used. Traditional streetlights, such as those using high-pressure sodium (HPS) or mercury vapor lamps, typically consume between 150 to 400 watts per light. Modern LED streetlights, which are more energy-efficient, consume between 30 to 150 watts while providing equivalent or better illumination. The total energy consumption for street lighting in a city can range from several thousand to millions of kilowatt-hours (kWh) annually, depending on the number of streetlights, their operating hours, and the power ratings of the luminaires. For instance, in large metropolitan areas, street lighting can account for 30-50% of the total energy usage of municipal services. To reduce consumption, many municipalities are adopting smart lighting systems and renewable energy solutions, such as solar-powered streetlights, which significantly cut down on grid dependency and operational costs while aligning with sustainability goals. Accurate assessment and optimization of power usage in street lighting are crucial for energy efficiency and cost management.

An occupancy sensor is a device designed to detect the presence or absence of people within a specific area. By leveraging technologies such as infrared, ultrasonic, microwave, or a combination of these, it can sense motion, body heat, or sound to determine whether a space is occupied.

Automation is common usage for these sensors, especially in security, HVAC, and lighting systems. To increase convenience and energy economy, they can, for example, automatically turn lights on when someone enters a room and off when the room is empty. By making sure areas are lit when necessary or warning systems of an unauthorized presence, occupancy sensors improve safety and security in addition to saving energy.

The use of this smart technology in streetlights can reduce the amount of electrical energy wasted. Occupancy sensors are essential for resource optimization and user experience enhancement in a world that is embracing smarter and more sustainable solutions.

This project focuses on designing and implementing a Smart Streetlighting System utilizing occupancy sensors to optimize energy consumption. The proposed system automatically adjusts the lighting intensity based on the presence of pedestrians or vehicles, ensuring that energy is used only when necessary. By integrating advanced sensor technology with modern lighting systems, the project aims to provide a cost-effective and environmentally friendly alternative to conventional streetlights.

The key objectives of this project include:

1. Reducing energy wastage by activating lights only in occupied areas.
2. Enhancing the lifespan of streetlights through optimized usage.
3. Improving safety and visibility in public spaces without compromising on efficiency.

The report outlines the measurement of the light intensity of the streetlights, the development process of the occupancy sensor circuit, including the hardware and software components used, the methodology adopted, and the results obtained. This innovative approach demonstrates the potential of smart technologies in transforming traditional systems to meet contemporary challenges.

Chapter 2: Literature Review

1. X Guo et al. (2010) investigated the performance of occupancy-based lighting control systems, which utilize sensors to detect occupancy and adjust lighting automatically. He found that the PIR sensors detect the change in temperature patterns within their field of view without emitting energy themselves. The sensors have Fresnel lens. The lens divides the field into zones, enhancing detection accuracy for movement. He also found that they are most effective for body heat at ~10 μm wavelength and are sensitive to movements at specific ranges:

   Hand movements: ~4.5 m (15 ft)

   Arm/torso movements: ~6 m (20 ft)

   Full body movements: ~12 m (40 ft)

   However, he observed that sensitivity of the PIR sensor decreases with distance due to zone widening and it needs a direct line of sight to detect movement effectively.

   He also found that Automated occupancy-based systems save more energy compared to manual switching, with estimated savings ranging from 15-75%. Proper installation and commissioning (e.g., sensor tuning, placement, and adjustments) are crucial to achieving these savings. He noted that shorter time delay settings (5-10 minutes) enhance savings but can reduce lamp life. His studies show varying results based on occupancy patterns, daylight availability, and system configurations, with average savings ranging from 20-30%

2. Richman et al. (1996) examined 141 sample spaces and found savings between 3%-50%, depending on time delay settings and space type, with greater potential in private offices and restrooms.

3. Floyd et al**.** (2010) reported up to 19% savings in office buildings but noted challenges such as false triggers and higher energy usage in some cases.

4. Maniccia et al. (2001) observed 43% savings in offices when PIR sensors were combined with building automation systems with a 30-minute time delay.

Chapter 3: Circuit Diagram and Working Principle

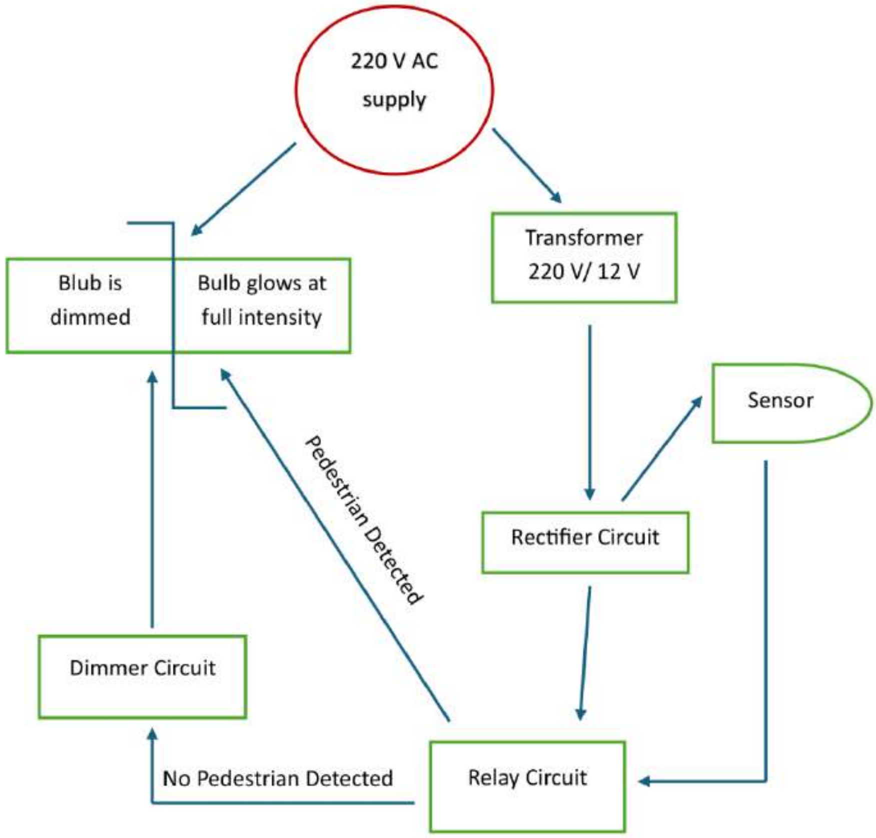


Fig 1: Block Diagram of the model

3.1 Methodology

The designing of our project "Smart streetlight technology using occupancy sensor" has been done by focusing on the development of the three following functions: Object detection, Signal amplification, and automated dimming.

1. Obstacle detection via IR sensor

For the detection of obstacles, we chose an infrared sensor. An IR sensor emits infrared light and detects reflected rays from any obstacle within its range. It then sends the signal to the base of the transistor.

2.Signal amplification and relay triggering

The transistor acts as a switch and gets triggered by the IR sensor's output.

When the transistor is on, it energizes the relay which then activates the relay to switch ON the 220V AC lamp at full brightness.

3. Automatic brightness control (dimming) When there is no object within the range of the sensor, the sensor output is low, and the transistor remains OFF and the relays are deactivated. In that case, instead of turning off the lamp completely, a phase angle control using TRIAC, DIAC, Resistors, potentiometer and capacitor, the lamp's brightness is controlled and kept in a dimmed state.

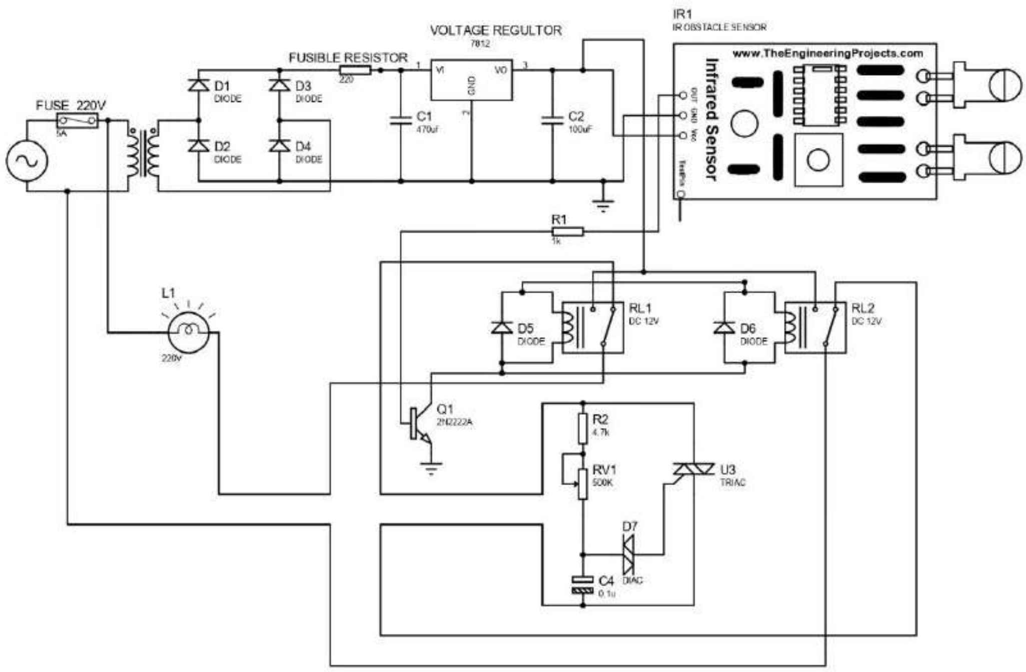


Fig 2: Circuit diagram of the occupancy sensor-based streetlight system

3.2 Working Principle

The power supply section:

- To provide a stable DC power supply (12V) to the IR sensor and control components, we employed the following:

- A 220V AC main is stepped down using a 220/12 V AC transformer at 50Hz.
- A bridge rectifier (D1, D2, D3 & D4) converts this 12 Volt AC to pulsating DC.
- Smoothing Capacitors (C1, C2) filter the near DC to reduce the ripple.
- A 7812 Voltage regulator IC outputs a constant 12V DC.
- This 12V powers the PIR sensor, transistor and relays.

Object Detection using IR sensor

- The PIR sensor emits infrared light and detects the reflection from nearby objects.
- When no object is present, the sensor output is LOW.
- When an object is detected, the sensor output is HIGH.
- This output is then fed into a transistor (2N2222A) to control further switching.
- 2N2222A is used as it can handle maximum current of 800ma (the circuit uses two relays consuming 110ma each).

Switching Mechanism using transistor and relays

- When the sensor detects an object, the base of Q1 transistor (2N2222A) receives current.
- Q1 conducts and acts like a switch, energizing relay RL1 and relay RL2.
- The energizing of relays connects full 220V AC to the streetlight (L1).
- The lamp turns ON at full brightness.

Dimming Mechanisms using TRIAC-DIAC Phase Control

- When no object is detected, the IR sensor does not trigger Q1 (RL1 and RL2 remain off).
- Instead of Switching off the lamp, a phase control dimmer circuit involving TRIAC(U3), DIAC(D7), Resistors (R2, RV1), and Capacitor (C4) controls the firing angle of the AC waveform.
- This means the lamp only receives part of the AC cycle, so it glows dimly.
- The RV1(variable resistor) allows us to adjust how dim the lamp should be.

Safety and protection

- Diodes D5 and D6 (Flyback Diodes) protect against back EMF from relays.
- The 5A/250v AC fuse and the fusible resistor (220ohms) protects the entire circuit from overcurrent and short circuit.

Chapter 4 Hardware Requirements

4.1 220V AC TO 12V AC Transformer

A 220/12V AC transformer is a step-down transformer commonly used in electronic power supplies, particularly when a lower voltage output is needed. The designation "220/12V AC" refers to the voltages at the primary and secondary windings of the transformer respectively. It means that the transformer has a single primary winding, with two output terminals: one at each end of the secondary winding.

Purpose in the model

To provide the model with a proper power supply, we used it to step down the 220V AC supply to 12V AC.

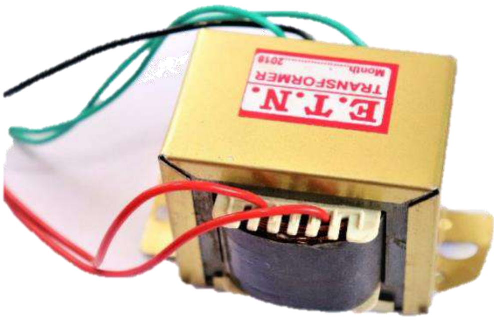


Fig 3: 220V AC to 12V AC Transformer

4.2 5A, 250V Fuse

A 5A, 250V fuse is an electrical safety device designed to protect circuits and components from damage caused by excessive current. The ratings "5A" and "250V" indicate that the fuse is designed to allow continuous current flow of up to 5 A and 250V under normal operating conditions.

Purpose in Model

To keep the rectification circuit, sensor circuit, and dimming circuit from any overcurrent, short circuit orovervoltage that may lead to damaging of the components.

Fig 4: 5A, 250V Fuse

4.3 IN4007 Diode

The IN4007 Diode is a widely used general-purpose rectifier diode. It belongs to the IN400x series of diodes, which are commonly used in power supply circuits. The IN4007 is particularly favored for its high voltage rating- it can withstand a reverse voltage of up to 1000 Volts and conduct up to 1 Ampere of continuous forward current.

Purpose in Model

It is used for:

- Making of bridge rectifier circuit to convert the 12V AC to 12 V pulsating DC.
- Protection against back EMF from the relays.

Fig 5: IN4007 Diode

4.4 7812 IC

The 7812 IC is a fixed voltage regulator from the widely used 78xx series, specifically designed to provide a stable 12V DC output. Like other regulators in the series, the 7812 is a three-terminal device with pins labelled Input (Vin), Ground (GND), and output (Vout). It accepts an input voltage typically in the range of 14V to 35V and regulates it down to a constant 12V, making it suitable for powering devices and circuits that require a steady 12V supply.

Purpose in the model

The input to the 7812 IC is an unregulated DC voltage coming from the bridge rectifier and the filter capacitors. The IC regulates it and provides a constant 12V DC regardless of the input voltage (within specified limitations).

Fig 6: 7812 IC

4.5 PIR Sensor

A PIR (Passive Infrared) sensor is an electronic device used to detect motion by sensing infrared radiation emitted by warm objects such as humans and animals. Unlike active sensors, which emit energy to detect objects, PIR sensors are called "passive" because they do not generate any radiation themselves; they simply receive and interpret infrared signals present in their surroundings

Purpose in the model

The PIR sensor detects the motion of living beings and objects with heat and when motion is detected it sends a HIGH signal to the transistor. The two variable resistors on the PIR sensor is to adjust the sensitivity and the delay at which the output remains in HIGH state after the PIR sensor stops detecting.

Fig 7: PIR Sensor

4.6 2N2222A Transistor

The 2N2222A is a NPN Bipolar Junction Transistor (BJT) used as a switching and amplification in electronic circuits. It has three terminals: base, collector and emitter. When a small current flows into the base, it allows a much larger current to flow from collector to emitter.

Purpose in the model

In the model, the 2N2222A transistor acts as an intermediary switch between the PIR sensor and the relay. When the PIR sensor detects motion and sends HIGH signal, a voltage is applied to the base of the transistor, allowing current to flow from the collector to the emitter. This activates the relay coil, which is connected to the transistor’s collector.

Fig 8: 2N2222A Transistor

4.7 Relay

A relay is an electromechanical switching device that uses a 12V DC electrical signal to control the opening and closing of one or more sets of electrical contacts. It can withstand a maximum current of 10A. It acts as an interface between low power control circuits and high-power electrical loads. The relay we used in our model is a Single Pole Double Throw (SPDT) relay. The relay has three terminals called Common, Normally Closed (NC) and Normally Open (NO).

Purpose in the model

To control the AC powered streetlight using the DC controlled circuit (sensor and transistor). When the sensor detects motion and the relay is activated via a signal from the transistor, the contacts of relay connect the 220V AC supply to the streetlight, turning it ON at full brightness. When the sensor do not detect any motion, the contacts of the relay remain open.

Fig 9: Relay

4.8 DIAC

A DIAC (Diode for Alternating Current) is a bidirectional semiconductor device that is commonly used to trigger TRIACs in power control applications, such as light dimmers, motor speed controllers, and heater controls. Unlike standard diodes or transistors, the DIAC does not conduct current until the voltage across it reaches a certain breakover voltage, typically in the range of 30 to 40 Volts. Once this voltage is exceeded, the DIAC switches to a low resistance conductive state, allowing current to flow in either direction. This characteristic makes it symmetrical and bidirectional, meaning it behaves the same way regardless of the polarity of the voltage applied. The DIAC is usually used in phase control circuits to provide a sharp triggering pulse to the gate of a TRIAC, enabling accurate and consistent control of AC power. It helps to improve the triggering reliability of the TRIAC, especially in situations where the input voltage waveform may be noisy or irregular.

Purpose in the model

When no object is detected by the sensor, the lamp should not be fully ON but should stay dim to save energy. The purpose of the DIAC is to trigger the TRIAC at the right moment in each AC cycle, enabling phase angle control for dimming the streetlight, save energy, reduce the flickering of the lightbulb, and to avoid the complete blackout of the bulb in the case that a fast moving object (example: a fast moving vehicle) enters and leaves the range of the sensor in an instant. Since the DIAC only conducts at a certain voltage, it ensures that the TRIAC gets sharp, timed, triggering pulse for stable dimming.

Fig 10: DIAC

4.9 TRIAC

A TRIAC (Triode for Alternating Current) is a three-terminal semiconductor device that is widely used for controlling power in AC circuits. It can conduct current in both directions, making it especially suitable for alternating current applications like light dimmers, fan speed controllers, and smart lighting systems. The three terminals of a TRIAC are Main Terminal 1 (MT1), Main Terminal 2 (MT2), and the Gate. When a small triggering current is applied to the gate, the TRIAC turns on and allows current to flow between MT1 and MT2, regardless of the current direction. Once turned on, the TRIAC remains conducting until the current flowing through it drops below a certain threshold known as the holding current, which usually happens at the end of each AC half-cycle.

One of the most important features of a TRIAC is its ability to be triggered by both positive and negative gate pulses, offering greater flexibility in AC control. This characteristic allows it to be used in phase angle control, where it switches on at a specific point in each AC cycle to adjust the amount of power delivered to a load, such as a streetlight. This is essential for dimming functions in smart streetlight systems. However, TRIACs can be sensitive to rapid changes in voltage (dv/dt), which may unintentionally trigger the device. To prevent this, snubber circuits are often used alongside TRIACs. In summary, the TRIAC is a compact, efficient, and reliable component that plays a vital role in modern AC power control applications.

Purpose in the model

The TRIAC is used to control the amount of AC power delivered to the streetlight bulb. This is done through a method called phase angle control, where the TRIAC is triggered to turn on at a certain point during each half-cycle of the AC waveform. The later in the cycle the TRIAC is triggered, the less power is delivered to the lamp, resulting in a dimmer light output.

When the IR or PIR sensor does not detect motion, the control circuit does not energize the relay for full brightness. Instead, the TRIAC circuit takes over. A timing circuit involving a capacitor and resistors charges up with each AC cycle. When the voltage across the capacitor reaches the breakover point of the DIAC, the DIAC conducts and sends a pulse to the gate of the TRIAC, triggering it. This setup allows the TRIAC to conduct only part of the AC waveform, effectively reducing the power delivered to the lamp. As a result, the bulb operates in a dimmed state, which is sufficient for low activity periods and significantly improves energy efficiency.

Fig 11: TRIAC

4.10 Luxmeter

We must measure light in many tasks. For this, the lux meter should be used. To measure light accurately, a lux meter is a very suitable device. It is calibrated to read directly in lux. Lux is a unit of illumination which is in the International System of Units that is SI. One lux Latin for “light” is said to be the amount of illumination that is provided when one lumen is evenly distributed over an area of one square meter. This is also said to be equivalent to the illumination that generally exists on the surface, all points of which are one meter from a point source of one international candle or candela.

One lux is said to be equal to 0.0929 foot-candle. The meters measure illumination in terms of lux or foot-candles are known as Lux meters or Light meters. A lux is equal to the total intensity of light that falls on a one square meter surface that is one foot away from the point of source of light. A foot-candle is equal to the total intensity of light that falls on a one square foot surface that is one foot away from the point source of light.

Purpose in the Model

To measure the illuminance when the sensor detects motions and the bulb is at ON state and measure the illumination when the sensor does not detect any motion and the bulb is at dim state.

Fig 12: Luxmeter

Chapter 5: Results and Discussion

5.1 Working Principle

The proposed system for smart technology using occupancy sensors was developed using transformers, PIR sensors, relays, TRIAC, DIAC, and other associated components. The model was tested under different conditions, and the sensors response to infrared heat from human bodies, animals, and vehicles was successfully recorded.

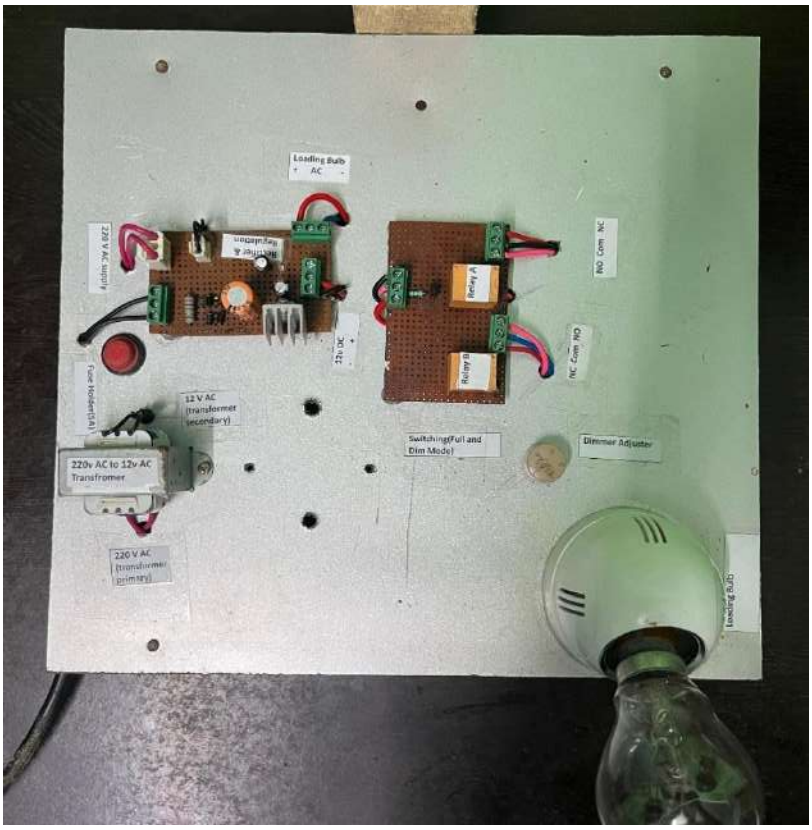


Fig 13: Top view of the developed model

Fig 14: PIR Sensor

Fig 15: Rectifier, protection and regulation section

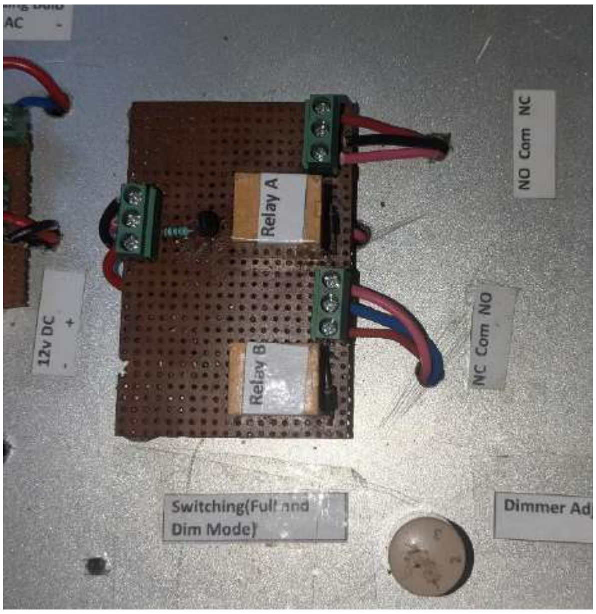


Fig 16: Switching Section (Full and Dimmer Mode)

Fig 17: Dimmer Adjuster

Fig 18: Back view (Dimmer Circuit)

5.2 Observations from the model

In the absence of a person or animal within the range of the sensor, we saw that the model displayed different values of voltage, current and illuminance from the bulb compared to when a person or vehicle was present within the sensor's range.

The observed readings are as shown in the following table:

Sl no.	Loading Lamp	Obstacle	Bulb State	Wattage	Current	Voltage (AC)	Illuminance
1	100W	Present	ON	100	420 mA	239.4 V	27 Lux
2	100W	Absent	Dim	23	185 mA	126 V	9 Lux
3	12W	Present	ON	12	49mA	238.9V	120 Lux
4	12W	Absent	Dim	3	27mA	112.6V	36 Lux
5	9W	Present	ON	9	39mA	225.8V	99 Lux
6	9W	Absent	Dim	2.45	20mA	119.8V	27 Lux

Table 1: Observations from the circuit model

From the above observations, we can infer that during periods of inactivity, the model had reduced the illuminance of the bulb to 33% of its full value. This is due to the arrangement of the PIR sensor and the control circuit of the model. The advantage of this is that the wastage of energy is minimized and light pollution is reduced as well

5.3 Luxmeter readings of streetlights in NERIST Campus

Three different locations of NERIST campus were chosen for finding the illuminance of the street lighting.

Case 1: Near AE Department

Case 2: Near EE Department

Case 3: In front of ME Department

For the measurements, 24x5 m surface area was taken on which square grids of 1X1 m each were plotted. The total number of square grids was 100. In each square grid, the illuminance of the streetlights present in the area were measured using luxmeter. So, for each location there are 100 values of illuminance.

Now using these values, we have obtained the Isolux diagram using MATLAB simulation.

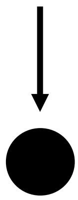
Here are the observations of each of the locations along with their Isolux diagram:

Case 1: In front of the Agriculture Engineering Department

Fig 19: Streetlight in front of Agriculture Engineering Department

Table 2: Luxmeter data of 24x5 meter area in front of Agriculture Engineering Department

Source of light



Unit- lux

11.1	10.8	11.3	11.1	10.3
12.3	12.5	12	11.6	11
13.7	13.5	13	12.6	11.6
15.1	14.8	14.3	13.8	13.4
16.4	15.6	16.2	15.2	14.3
18.5	19.2	17.7	16.9	14.3
21.1	21.0	19.1	18	15.1
23.1	21.9	21	19.3	15.7
23.4	24.2	22.6	20.5	16.9
24.6	23.4	23.5	21.4	18.3
26	24.8	23.5	21.7	18.4
27.5	27.2	24.9	21.6	18.4
28.8	27.6	25.6	22.8	18.6
28.9	27.8	25.9	23.3	18.6
26.8	26.9	25.5	22.8	19.8
27.1	27.5	25.6	22.4	19.3
25.9	23.3	22.4	20.4	18.6
24.5	24.8	22.1	19.8	13.2
22.6	21.8	20	18.2	16.1
19.2	19.8	18.6	16.8	15.4
17.2	18.2	16.9	15.6	14.6
16.6	15.8	14.8	14.1	13.5
14.5	14.1	13.1	12.8	12.1
12.7	13.2	12.4	10.1	9

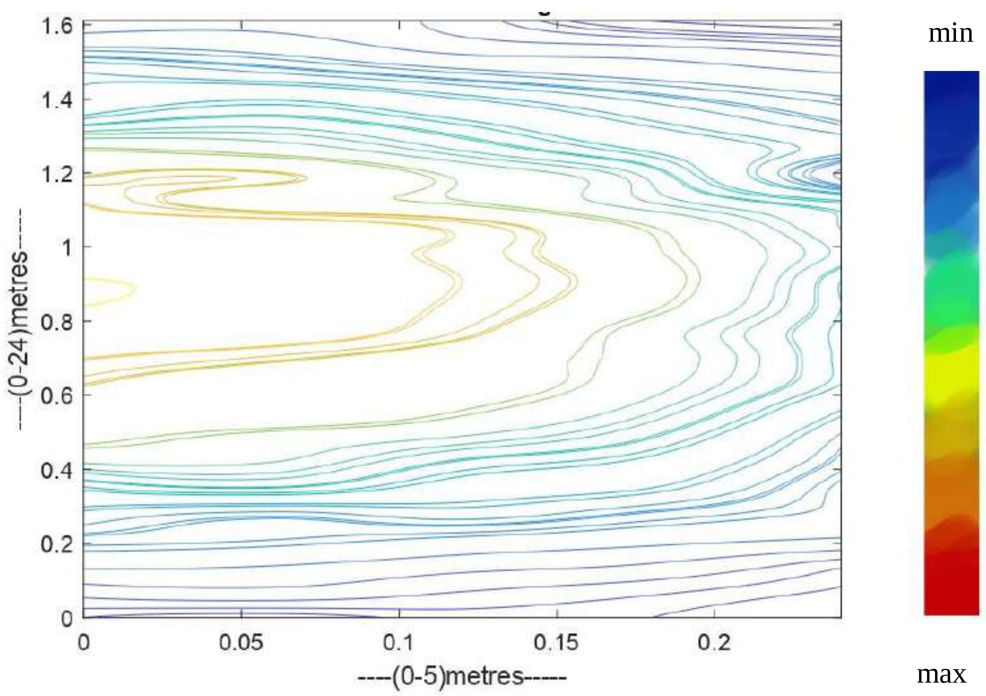


Fig 20: Isolux plot of luxmeter data gathered in front of Agriculture Engineering Department

Case 2: In front of the Electrical Engineering Department

Fig 21: Streetlight in front of the Electrical Engineering Department

Table 3: Luxmeter data of 24x5 meter area in front of the Electrical Engineering Department

Unit- lux

Source of light

18.9	20	25	24.4	28
21.7	22	28.1	26.3	31
24.2	26	30.2	29.8	31.4
27.5	22	32	30	33.9
30.8	30	34.9	32	36
33.4	26	35.6	35	37
34.1	28.3	37.8	36.1	42
35.8	40	41.2	37.2	46.5
39.6	34.3	44.9	40.9	46.8
41.1	36.8	46.2	43.2	46.3
41.8	47.7	46	44.4	46.4
41	47.1	43	46	45
36.9	40.1	41	43.6	43
36.4	42	40.3	40.9	40.7
35.9	40.3	33	38.7	40
34.6	35.2	30.6	36.2	39
32.2	35.4	29.2	34.5	36
30.9	31.7	25.3	33.8	33
27.8	26.4	25.1	32.5	27
27.8	25.4	24	29.4	25
27.2	23.6	22.6	28.1	22
24.7	28.8	21.2	28.3	20
21.6	24.3	22.8	26	18.5
17.5	20	18.4	25	17.9

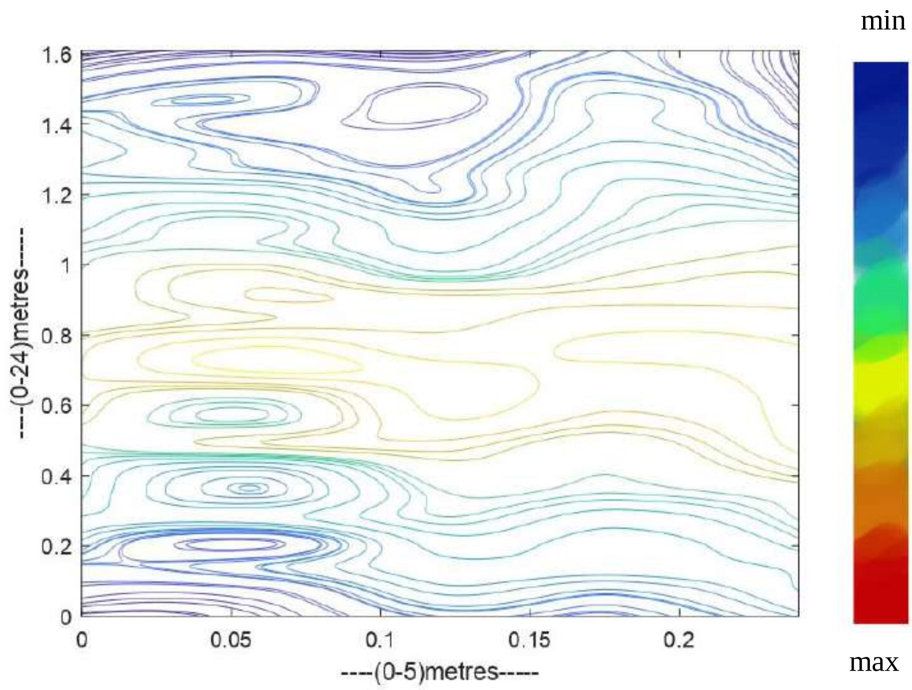


Fig 22: Isolux plot of luxmeter data gathered in front of the Electrical Engineering Department

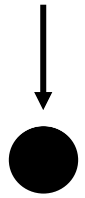
Case 3: In front of the Mechanical Engineering Department

Fig 23: Streetlight in front of the Department of Mechanical Engineering

Table 4: Luxmeter data of 24x5 meters in front of the Mechanical Engineering Department

Unit-lux

Source of light



14.4	16.4	15.8	14.1	24.2
16.5	18.9	18.5	15.9	23.8
19.8	22.1	20.7	17.3	22.6
23.3	24.8	23.2	20.6	23.1
26.6	27.4	25.6	22.1	22.6
28.8	29.5	26	23.3	21.8
29.9	30.3	26.2	23.4	21.1
30.2	28.5	26.6	24.6	21.9
30.4	29	27.2	25.6	20.4
30.8	29.6	28.1	26.2	17.4
32.4	28.8	27.6	26.5	16.4
34	30.8	28.8	26.4	14.8
29.7	24.9	25	24.1	23.3
29.6	25.3	24.5	23.8	21.6
22.1	23.4	23.9	23.1	22.7
26.4	20.8	23.2	22.8	21.5
18.9	18.6	21.6	20.9	20.5
15.5	15.8	19.2	18.5	18.5
13.8	13.8	16.5	15.9	15.9
12.4	11	13.8	13.3	12.9
9.7	9.9	11.7	11.3	10.9
10.3	8.2	9.7	9.3	9.5
6.6	7.1	8.2	8.1	7.8
7.1	6.1	7.1	6.9	6.7

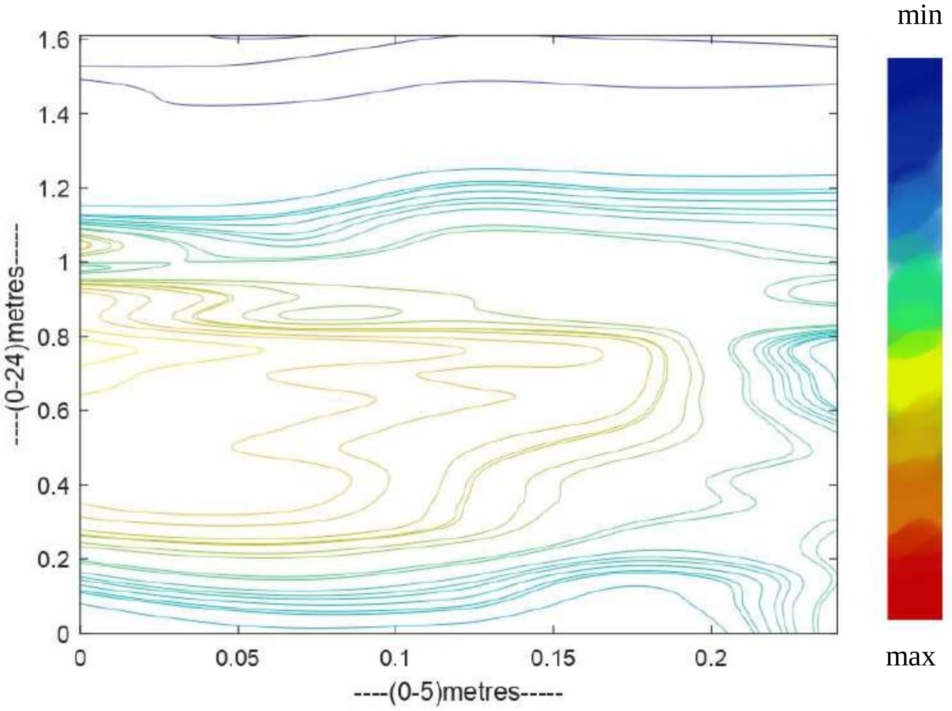


Fig 24: Isolux plot of luxmeter data gathered in front of the Mechanical Engineering Department

All the above luxmeter measurements were taken from traditional streetlights that always operate at full illuminating intensity during their operating hours.

Chapter 6: Future Works

If the proposed PIR sensor-based street lighting system is implemented across the NERIST campus, it can contribute significantly to energy conservation and the adoption of smart infrastructure. Based on our prototype testing, the model can reduce the illuminance (lux) level by approximately 33% when no motion is detected. This dimming effect, when applied during nighttime hours in the absence of pedestrians or vehicles, would directly result in a reduction in power consumption by a similar proportion, as the electrical energy required to maintain dim lighting is substantially lower than that for full brightness.

Given that large portions of the campus often remain unoccupied during late-night hours, this system can help avoid unnecessary energy usage while still maintaining a basic level of illumination for safety and visibility. The light automatically returns to full intensity the moment motion is detected by the PIR sensor, ensuring that there is no compromise on visibility or security for pedestrians walking through the area.

In addition to energy savings, the system can lead to lower electricity bills for the institute and extend the lifespan of lighting equipment, since operating lights at lower intensities reduces wear and thermal stress on the components.

Chapter 7: Conclusion

In this project, we developed a PIR sensor-based automatic street lighting system designed to reduce energy consumption by dimming streetlights when no motion is detected and restoring full brightness upon detecting movement. The system employs a relay-based control mechanism integrated with a TRIAC-based dimmer circuit to switch between two brightness levels. The core objective is to maintain adequate visibility while minimizing unnecessary power usage during low-traffic periods.

To understand the lighting currently in place, we measured the lux levels at three different streetlight locations within the NERIST campus using a calibrated lux meter. These readings helped us establish a baseline for illumination levels typically provided by existing streetlights. Separately, we tested our prototype under controlled conditions and found that it can reduce the lux level by approximately 33% when operating in dim mode.

Although our system was not physically implemented in the existing campus infrastructure, the controlled testing results indicate that if deployed, it could significantly contribute to energy conservation by lowering the power delivered to streetlights during periods of inactivity. This would not only reduce electricity consumption but could also extend the lifespan of the lighting equipment due to reduced thermal and electrical stress.

In conclusion, the results of our prototype testing suggest that integrating this intelligent lighting control system into NERIST’s street lighting network could offer a practical and sustainable approach to reducing energy usage while maintaining necessary illumination levels for safety and security.

BIODATA OF PROJECT TEAM MEMBERS

1. **Nending Tagia**

   Roll Number: D/20/EE/108
   Email: nendingtagia007@gmail.com
   Joined NERIST: 2020, through NEE II
   Project Contributions: Primarily focused on report writing and drafted detailed project reports.

2. **Vekhoyi Nyekha**

   Roll Number: D/21/EE/113
   Email: unknwnnyekha@gmail.com
   Joined NERIST: 2021, through JEE
   Project Contributions: Played a key role in hardware development and ensured proper functionality of hardware components and refinement of project designs

3. **Tezenmo Woch**

   Roll Number: D/21/EE/012
   Email: tezenmowoch2000@gmail.com
   Joined NERIST: 2019, through NEE I
   Project Contributions: Focused on hardware development and ensured accurate simulations were carried out.

4. **Dhrubajyoti Sonowal**

   Roll Number: D/21/EE/002
   Email: sonowaldhruba00@gmail.com
   Joined NERIST: 2019, through NEE I
   Project Contributions: Focused mainly on simulation of the project and contributed to design and editing tasks.

References

[1] X. Guo, D.K. Tiller, G.P. Henze, C.E. Waters, “The performance of occupancy-based lighting control systems”, 2010.

[2] Meistad, Torill. (2014), “How energy efficient office buildings challenge and contribute to usability. Smart and Sustainable Built Environment.”

[3] Yuan Jin, “Lighting system control in office building using occupancy prediction based on historical occupied ratio” 2019.

[4] https://task61.iea-shc.org/Data/Sites/1/publications/IEA-SHC-Task61--Technical-Report-D1-Literature-Review.pdf

[5] http://lrt.sagepub.com/content/42/4/415

[6] https://www.lrc.rpi.edu/programs/nlpip/pdf/view/sros2.pdf

[7] https://www.scribd.com/document/455956043/LUX-METER- docx

[8] https://in.mathworks.com/?s_tid=gn_logo

[9] https://hirosarts.com/blog/

[10] https://intl.lutron.com/asia/Education-Training/Pages/lce/greenbenefits.aspx

[11] https://www.energy.gov/femp/articles/wireless-occupancy-sensors-lighting-controls-applications-guide-federal-facility

[12] https://beei.org/index.php/EEI/article/view/1527

Appendices

1. MATLAB code for generating Isolux plot of streetlights in front of AE department

```
grid_x=0:0.06:0.24;
grid_y=0:0.07:1.61;
val_lux=[
11.1    10.8    11.3    11.1    10.3
12.3    12.5    12      11.6    11
13.7    13.5    13      12.6    11.6
15.1    14.8    14.3    13.8    13.4
16.4    15.6    16.2    15.2    14.3
18.5    19.2    17.7    16.9    14.3
21.1    21.0    19.1    18      15.1
23.1    21.9    21      19.3    15.7
23.4    24.2    22.6    20.5    16.9
24.6    23.4    23.5    21.4    18.3
26      24.8    23.5    21.7    18.4
27.5    27.2    24.9    21.6    18.4
28.8    27.6    25.6    22.8    18.6
28.9    27.8    25.9    23.3    18.6
26.8    26.9    25.5    22.8    19.8
27.1    27.5    25.6    22.4    19.3
25.9    23.3    22.4    20.4    18.6
24.5    24.8    22.1    19.8    13.2
22.6    21.8    20      18.2    16.1
19.2    19.8    18.6    16.8    15.4
17.2    18.2    16.9    15.6    14.6
16.6    15.8    14.8    14.1    13.5
14.5    14.1    13.1    12.8    12.1
12.7    13.2    12.4    10.1    9


]

%------------------interpolation
grid_x_int=0:0.06/100:0.24
grid_y_int=0:0.07/100:1.61
[grid_x_int,grid_y_int]=meshgrid(grid_x_int,grid_y_int);
val_lux_int=interp2(grid_x,grid_y,val_lux,grid_x_int,grid_y_int,'cubic');
```

```
v=[7.1 6.2 9.3 22.6 24.5 25.9 18.6 19.3 25 20.4 28.8 26 4.6 18.3 16.9 21 22.3 20
26.5 27.8 33 35.8 31 40.1 34.3]
any(isnan(val_lux_int))  % Check for NaN
any(isinf(val_lux_int))  % Check for Inf
min(val_lux_int(:))
max(val_lux_int(:))
contour(grid_x_int,grid_y_int,val_lux_int,v)
title('ISO LUX Streetlight in front of ME Department ')
xlabel('----(0-5)metres-----')
ylabel('----(0-24)metres-----')
```

2. MATLAB code for generating Isolux plot of streetlights in front of EE Department

```
grid_x=0:0.06:0.24;
grid_y=0:0.07:1.61;
val_lux=[
18.9    20      25      24.4    28
21.7    22      28.1    26.3    31
24.2    26      30.2    29.8    31.4
27.5    22      32      30      33.9
30.8    30      34.9    32      36
33.4    26      35.6    35      37
34.1    28.3    37.8    36.1    42
35.8    40      41.2    37.2    46.5
39.6    34.3    44.9    40.9    46.8
41.1    36.8    46.2    43.2    46.3
41.8    47.7    46      44.4    46.4
41      47.1    43      46      45
36.9    40.1    41      43.6    43
36.4    42      40.3    40.9    40.7
35.9    40.3    33      38.7    40
34.6    35.2    30.6    36.2    39
32.2    35.4    29.2    34.5    36
30.9    31.7    25.3    33.8    33
27.8    26.4    25.1    32.5    27
27.8    25.4    24      29.4    25
27.2    23.6    22.6    28.1    22
24.7    28.8    21.2    28.3    20
21.6    24.3    22.8    26      18.5
```

```
17.5	20	18.4	25	17.9



]

%------------------interpolation
grid_x_int=0:0.06/100:0.24
grid_y_int=0:0.07/100:1.61
[grid_x_int,grid_y_int]=meshgrid(grid_x_int,grid_y_int);
val_lux_int=interp2(grid_x,grid_y,val_lux,grid_x_int,grid_y_int,'cubic');


v=[7.1 6.2 9.3 22.6 24.5 25.9 18.6 19.3 25 20.4 28.8 26 4.6 18.3 16.9 21 22.3 20
26.5 27.8 33 35.8 31 40.1 34.3]
any(isnan(val_lux_int))  % Check for NaN
any(isinf(val_lux_int))  % Check for Inf
min(val_lux_int(:))
max(val_lux_int(:))
contour(grid_x_int,grid_y_int,val_lux_int,v)
title('ISO LUX Streetlight in front of ME Department ')
xlabel('----(0-5)metres-----')
ylabel('----(0-24)metres-----')
```

3. MATLAB code for generating Isolux plot of streetlights in front of ME department

```
grid_x=0:0.06:0.24;
grid_y=0:0.07:1.61;
val_lux=[
14.4	16.4	15.8	14.1	24.2
16.5	18.9	18.5	15.9	23.8
19.8	22.1	20.7	17.3	22.6
23.3	24.8	23.2	20.6	23.1
26.6	27.4	25.6	22.1	22.6
28.8	29.5	26	23.3	21.8
29.9	30.3	26.2	23.4	21.1
30.2	28.5	26.6	24.6	21.9
30.4	29	27.2	25.6	20.4
30.8	29.6	28.1	26.2	17.4
32.4	28.8	27.6	26.5	16.4
```

```
34      30.8    28.8    26.4    14.8
29.7    24.9    25      24.1    23.3
29.6    25.3    24.5    23.8    21.6
22.1    23.4    23.9    23.1    22.7
26.4    20.8    23.2    22.8    21.5
18.9    18.6    21.6    20.9    20.5
15.5    15.8    19.2    18.5    18.5
13.8    13.8    16.5    15.9    15.9
12.4    11      13.8    13.3    12.9
9.7     9.9     11.7    11.3    10.9
10.3    8.2     9.7     9.3     9.5
6.6     7.1     8.2     8.1     7.8
7.1     6.1     7.1     6.9     6.7



]

%------------------interpolation
grid_x_int=0:0.06/100:0.24
grid_y_int=0:0.07/100:1.61
[grid_x_int,grid_y_int]=meshgrid(grid_x_int,grid_y_int);
val_lux_int=interp2(grid_x,grid_y,val_lux,grid_x_int,grid_y_int,'cubic');


v=[7.1 6.2 9.3 22.6 24.5 25.9 18.6 19.3 25 20.4 28.8 26 4.6 18.3 16.9 21 22.3 20
26.5 27.8 33 35.8 31 40.1 34.3]
any(isnan(val_lux_int))  % Check for NaN
any(isinf(val_lux_int))  % Check for Inf
min(val_lux_int(:))
max(val_lux_int(:))
contour(grid_x_int,grid_y_int,val_lux_int,v)
title('ISO LUX Streetlight in front of ME Department ')
xlabel('----(0-5)metres-----')
ylabel('----(0-24)metres-----')
```